



# NORMANHURST BOYS HIGH SCHOOL

## MATHEMATICS EXTENSION 1

2016 HSC Course Assessment Task 4 (Trial Examination)

Friday August 12, 2016

### General instructions

- Working time – 2 hours.  
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

### SECTION I

- Mark your answers on the answer grid provided (on page 11)

### SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

**BOSTES NUMBER:** .....

**# BOOKLETS USED:** .....

**Class** (please ✓)

☐ 12M1 – Mr Sekaran

☐ 12M3 – Mr Lam

☐ 12M4 – Mr Wall

☐ 12M2 – Mrs Bhamra

☐ 12M5 – Mrs Gan

Marker's use only.

| QUESTION | 1-10               | 11                 | 12                 | 13                 | 14                 | Total              | % |
|----------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|---|
| MARKS    | $\frac{\quad}{10}$ | $\frac{\quad}{15}$ | $\frac{\quad}{15}$ | $\frac{\quad}{15}$ | $\frac{\quad}{15}$ | $\frac{\quad}{70}$ |   |

## Section I

10 marks

Attempt Question 1 to 10

Mark your answers on the answer grid provided (labelled as page 11).

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### Questions

### Marks

1. What is the number of asymptotes on the graph of  $y = \frac{1}{x^2 - 1}$ ? 1

(A) 1                      (B) 2                      (C) 3                      (D) 4

2.  $A$  is the point  $(-1, 4)$  and  $B$  is  $(9, -6)$ . Which of the following are coordinates of a point that divides  $AB$  internally in the ratio  $3 : 2$ ? 1

(A)  $(1, 3)$                       (B)  $(3, 1)$                       (C)  $(-2, 5)$                       (D)  $(5, -2)$

3. The equation of the normal to the parabola  $x^2 = 4ay$  at the variable point  $P(2ap, ap^2)$  is given by  $x + py = 2ap + ap^3$ . 1

How many different values of  $p$  are there such that the normal passes through the focus of the parabola?

(A) 0                      (B) 1                      (C) 2                      (D) 3

4. A curve has parametric equations  $x = t - 3$  and  $y = t^2 + 2$ . 1

What is the Cartesian equation of this curve?

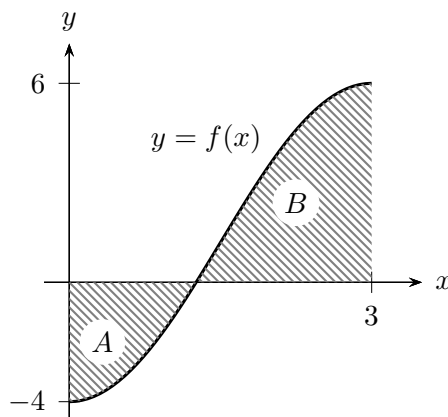
(A)  $y = x^2 - x - 1$                       (C)  $y = x^2 - 6x + 11$

(B)  $y = x^2 + x - 1$                       (D)  $y = x^2 + 6x + 11$

5. What is the value of  $\tan^{-1}\left(\tan \frac{5\pi}{4}\right)$ ? 1

(A)  $\frac{\pi}{4}$                       (B)  $-\frac{\pi}{4}$                       (C)  $\frac{5\pi}{4}$                       (D)  $-\frac{5\pi}{4}$

6. A graph of the function  $y = f(x)$  is shown. 1



Area  $A$  is equal to 3 square units, and area  $B$  is equal to 7 square units.

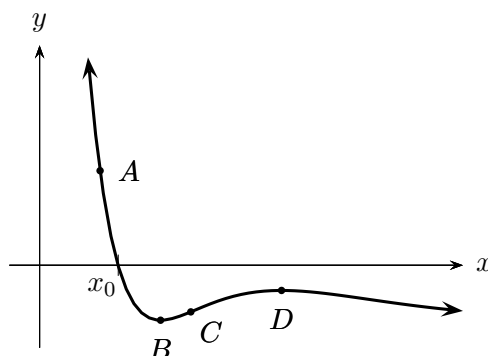
What is the value of  $\int_{-4}^6 f^{-1}(x) dx$ ?

(A) 4                      (B) 10                      (C) 14                      (D) 20

7. What is the largest value of  $m$  such that the equation  $\sin^{-1} x - mx = 0$  has three unique solutions? 1

(A)  $m = 0$                       (B)  $m = 1$                       (C)  $m = \frac{\pi}{2}$                       (D)  $m = \pi$

8.  $f(x)$  is shown in the following diagram, and  $f(x)$  has a root at  $x = x_0$ . Sahil is trying to find the first approximation to this root, from where a second approximation will be found by Newton's Method. 1

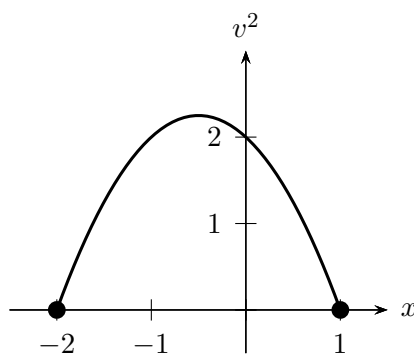


Which point should Sahil choose as his first approximation, so that his subsequent application of Newton's Method will produce a better approximation?

(A)  $A$                       (B)  $B$                       (C)  $C$                       (D)  $D$

9. A graph of the square of the velocity against displacement is shown.

1



What type of motion does this graph best describe?

- (A) Parabolic (C) Projectile  
(B) Exponential (D) Simple harmonic

10. Which of the following is *not* equal to  $\frac{d}{dx} \left( \frac{1}{2} v^2 \right)$ ?

1

- (A)  $v \frac{dv}{dx}$  (B)  $\frac{dv}{dt}$  (C)  $\ddot{x}$  (D)  $\dot{x}$

**Examination continues overleaf...**

## Section II

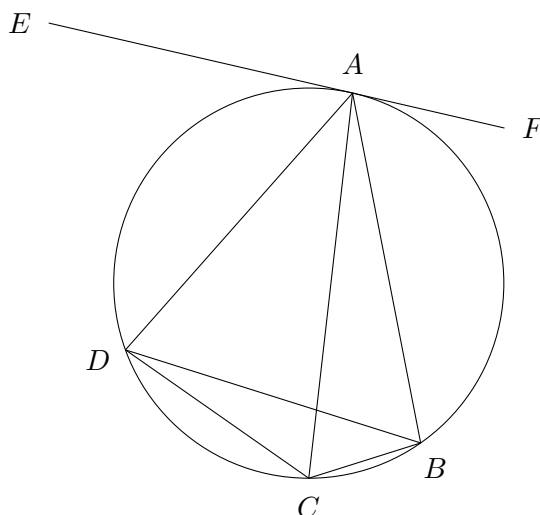
60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

| Question 11 (15 Marks)                                                                                                             | Commence a NEW booklet. | Marks |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------|
| (a) Find the acute angle between the lines $2x - y = 0$ and $x - 2y = 0$ , giving the answer correct to the nearest degree.        |                         | 2     |
| (b) Solve the inequality $\frac{x^2 - 4}{x} \geq 0$ .                                                                              |                         | 3     |
| (c) Find the roots of the equation $x^3 + 6x^2 - x - 30 = 0$ given one of the roots of the equation is the sum of the other roots. |                         | 3     |
| (d) $ABCD$ is a cyclic quadrilateral. $EF$ is a tangent at $A$ to the circle. $CA$ bisects $\angle BCD$ .                          |                         | 3     |

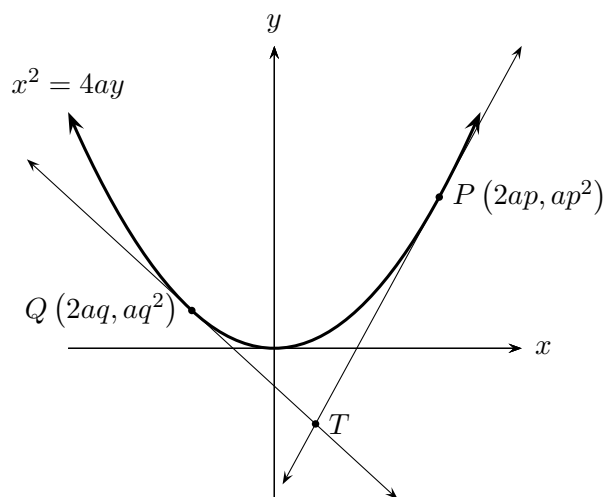


Show that  $EF \parallel DB$ .

Question 11 continued overleaf...

Question 11 continued from previous page...

- (e) The diagram shows the parabola  $x^2 = 4ay$ . The points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola. The tangents at  $P$  and  $Q$  intersect at the point  $T$ .



- i. By referring to the Reference Sheet or otherwise, show that  $T$  has coordinates  $(a(p+q), apq)$ . 2
- ii. The points  $P$  and  $Q$  move on the parabola such that the gradient of the chord  $PQ$  is always equal to  $\frac{a}{2}$ . 2

Show that the locus of  $T$  is parallel to the axis of the parabola.

**Examination continues overleaf...**

**Question 12** (15 Marks)

Commence a NEW booklet.

**Marks**

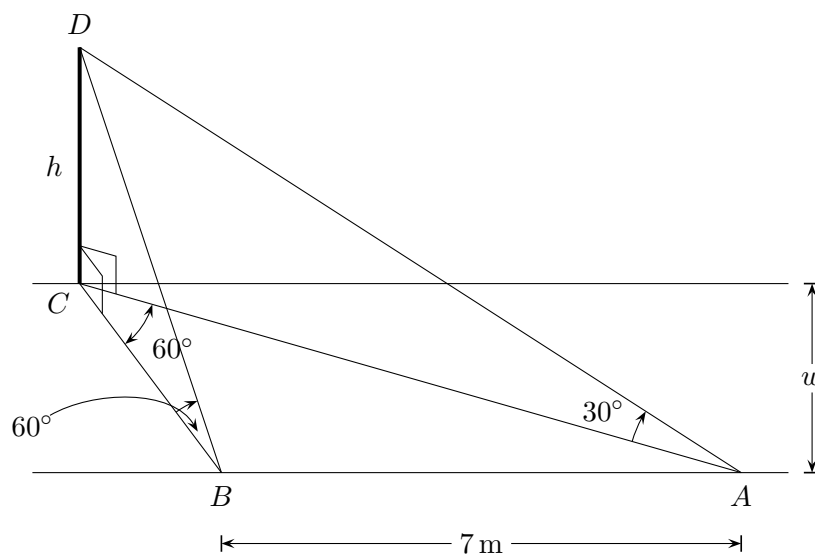
- (a) Prove by induction that:

**3**

$$\frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} + \cdots + \frac{2^n \times n}{(n+1)(n+2)} = \frac{2^{n+1}}{n+2} - 1$$

for all integers  $n \geq 1$ .

- (b) A footpath on horizontal ground has two parallel edges.  $CD$  is a vertical street sign pole of height  $h$  metres which stands with its base  $C$  on one edge of the footpath.  $A$  and  $B$  are two points on the other edge of the footpath such that  $AB = 7$  m and  $\angle ACB = 60^\circ$ . From  $A$  and  $B$ , the angles of elevation to the top of the pole at  $D$  are  $30^\circ$  and  $60^\circ$  respectively.



- i. Show that the exact height of the flagpole is  $h = \sqrt{21}$  metres. **3**
  - ii. By considering the area of  $\triangle ABC$ , find the exact width  $w$  of the footpath. **3**
- (c)
- i. Find the derivative of  $f(x) = \tan(x^2)$ . **2**
  - ii. Hence or otherwise, evaluate  $\int_{-a}^a x \sec^2(x^2) dx$ . **2**
- (d) Find the simplest expression for  $\sin(\cos^{-1}(x-1))$  in terms of  $x$  only. **2**

**Examination continues overleaf...**

**Question 13** (15 Marks) Commence a NEW booklet. **Marks**

(a) Evaluate  $\int \frac{1}{\sqrt{9-4x^2}} dx$ . **2**

(b) Use the substitution  $u = 3 + e^x$  to find the exact value of **3**

$$\int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx$$

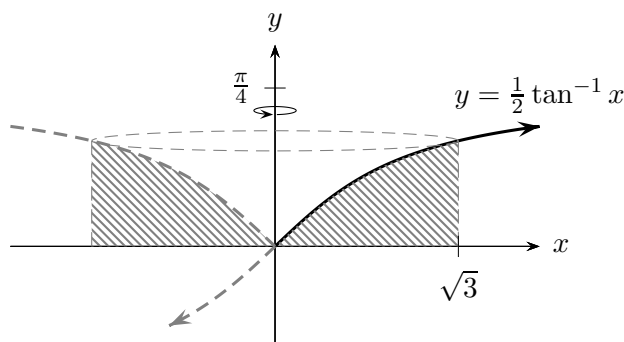
(c) i. Differentiate  $y = \tan^{-1}(e^x)$ . **2**

ii. Find the equation of the tangent to  $y = \tan^{-1}(e^x)$  at  $x = 0$ . **2**

iii. Discuss the behaviour of the curve as  $x \rightarrow \infty$ . **1**

(d) i. Evaluate  $\int \tan^2 2x dx$ . **2**

ii. Find the exact volume generated when the area between curve  $y = \frac{1}{2} \tan^{-1} x$ , the  $x$  axis between  $x = 0$  and  $x = \sqrt{3}$  is rotated about the  $y$  axis. **3**



(Extension 2 students: Do NOT use volumes by slicing or cylindrical shells)

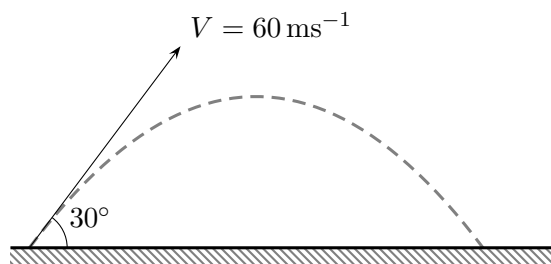
**Examination continues overleaf...**

**Question 14** (15 Marks)

Commence a NEW booklet.

**Marks**

- (a) A projectile is launched from the origin, across a level plain at  $30^\circ$  to the horizontal and at an initial speed of  $60 \text{ ms}^{-1}$ . Use  $g = 10 \text{ ms}^{-2}$ .



The displacement equations of motion are:

$$\begin{cases} x = 30\sqrt{3}t \\ y = 30t - 5t^2 \end{cases} \quad (\text{Do NOT prove these})$$

- i. Find the maximum height of the particle. **2**
  - ii. Find the velocity of the particle one second after launch. **2**
- (b) A particle is moving in simple harmonic motion with

$$\ddot{x} = -4x + 4$$

Initially, the particle is at the origin and is moving away from the origin with a speed of  $2\sqrt{3} \text{ ms}^{-1}$ .

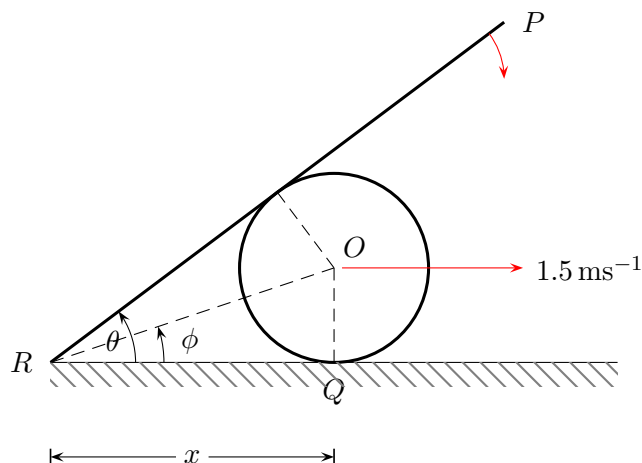
- i. Use integration to show that  $v^2 = -4(x - 3)(x + 1)$ . **2**
- ii. Find the centre and amplitude of the motion. **2**
- iii. Find the displacement-time equation for this particle. **3**

**Question 14 continued overleaf...**

Question 14 continued from previous page...

- (c) A rod  $RP$  is leaning on a disc of radius 1 m as shown in the diagram. The centre of the disc is labelled  $O$ .

One end of the rod  $R$ , is fixed to the ground, whilst the disc is tangential to the ground at  $Q$ .



The disc rolls to the right with a constant speed of  $1.5 \text{ ms}^{-1}$ . As the disc rolls, the other end of the rod  $P$  commences its descent towards the ground.

Let  $\angle ORQ = \phi$  and  $\angle PRQ = \theta$  and  $RQ = x$ .

- i. Write a relationship between  $\angle ORQ$ ,  $OQ$  and  $RQ$ . 1
- ii. Hence or otherwise, find the rate of change of  $\theta$  in radians per second, when  $x = 6 \text{ m}$ . 3

**End of paper.**

## Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

**BOSTES NUMBER:** .....

**Class** (please ✓)

☐ 12M1 – Mr Sekaran

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|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <b>1</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| <b>2</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| <b>3</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| <b>4</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| <b>5</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
| <b>6</b> –  | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |
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| <b>10</b> – | <input type="radio"/> A | <input type="radio"/> B | <input type="radio"/> C | <input type="radio"/> D |

## 2016 Mathematics Extension 1 HSC Course Trial Examination STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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2. Which topics did I need more help with, and what parts specifically?

- Q1-4, 8, 11 - Miscellaneous topics, Polynomials, Circle Geometry, Parametric Representation

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- Q5, 7, 12 - Induction, 3D Trigonometry, Inverse Trigonometry

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- Q6, 13 - Integration (of inverse trig/substitution/volumes), Curve Sketching.

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- Q9-10, 14 - Further applications of calculus to the physical world.

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3. What other parts from the feedback session can I take away to refine my solutions for future reference?

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## Sample Band 6 Responses

### Section I

1. (C) 2. (D) 3. (B) 4. (D) 5. (A)  
6. (C) 7. (C) 8. (A) 9. (D) 10. (D)

### Section II

#### Question 11 (Wall)

(a) (2 marks)

$$\begin{array}{l|l} 2x - y = 0 & x - 2y = 0 \\ y = 2x & y = \frac{1}{2}x \end{array}$$

$$\therefore m_1 = 2 \quad m_2 = \frac{1}{2}$$

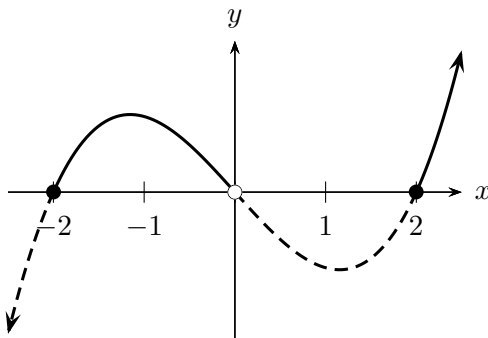
Apply angle between two lines formula,

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{2 - \frac{1}{2}}{1 + (2)\left(\frac{1}{2}\right)} \right| \\ &= \left| \frac{\frac{3}{2}}{2} \right| = \frac{3}{4} \\ \therefore \theta &= 36.869^\circ \dots \approx 37^\circ \end{aligned}$$

(b) (3 marks)

- ✓ [1] for multiplying by the square of the denominator.
- ✓ [1] for significant progress towards solution.
- ✓ [1] for removing  $x = 0$  as a possible solution.

$$\begin{aligned} \frac{x^2 - 4}{x \times x^2} &\geq \frac{0}{x^2} \\ x(x^2 - 4) &\geq 0 \\ x(x - 2)(x + 2) &\geq 0 \end{aligned}$$



$$\therefore -2 \leq x < 0 \text{ or } x \geq 2$$

(c) (3 marks)

- ✓ [1] for correct usage of elementary symmetric functions.
- ✓ [1] for significant progress towards solution.
- ✓ [1] for correct final solutions.

$$x^3 + 6x^2 - x - 30 = 0$$

Roots are  $\alpha$ ,  $\beta$  and  $\alpha + \beta$

- Sum of roots:

$$\begin{aligned} \alpha + \beta + (\alpha + \beta) &= -\frac{6}{1} = -6 \\ \therefore 2(\alpha + \beta) &= -6 \\ \alpha + \beta &= -3 \end{aligned}$$

- Product:

$$\begin{aligned} \alpha\beta(\alpha + \beta) &= -\frac{d}{a} = 30 \\ \alpha\beta(-3) &= 30 \\ \therefore \alpha\beta &= -10 \end{aligned}$$

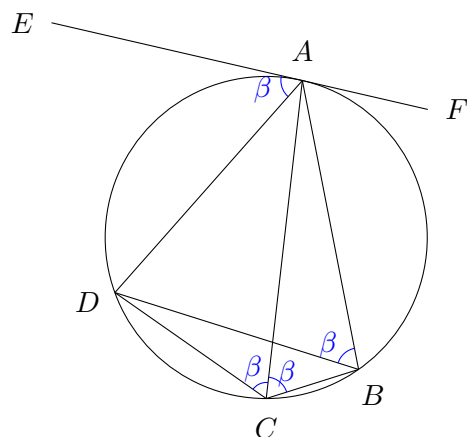
Form a quadratic with sum of roots  $-3$  and product of roots  $-10$ :

$$\begin{aligned} x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \\ x^2 + 3x - 10 &= 0 \\ (x + 5)(x - 2) &= 0 \\ \therefore x &= 2, -5 \end{aligned}$$

Hence roots are  $x = 2, -5, -3$ .

(d) (3 marks)

- ✓ [1] for appropriate usage of  $\angle$  in the alternate segment.
- ✓ [1] for appropriate usage of  $\angle$  in the same segment.
- ✓ [1] for final justification.

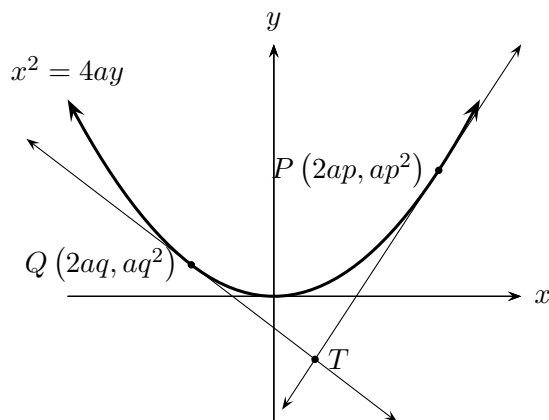


To prove  $EF \parallel DB$ : let  $\angle BAF = \alpha$  and  $\angle BCA = \angle ACD = \beta$ .

- Let  $\angle ACD = \beta$ .  
Then  $\angle ACB = \beta$   
( $CA$  bisects  $\angle BCD$ , given)
- $\angle ABD = \angle ACD = \beta$   
( $\angle$  in the same segment, on chord  $AD$ ).
- $\angle FAB = \angle ACB = \beta$   
( $\angle$  in the alternate segment)
- $\therefore EF \parallel DB$   
(alternate angles  $\angle FAB = \angle ABD$ )

(e) i. (2 marks)

- ✓ [1] for adequate steps to show  $x = a(p + q)$ .
- ✓ [1] for adequate steps to show  $y = apq$ .



Solving simultaneously for equation

of tangent at  $P$  and  $Q$ 

$$\begin{cases} y = px - ap^2 \\ y = qx - aq^2 \end{cases}$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$\therefore x = a(p + q)$$

Substitute into  $y = px - ap^2$ :

$$\begin{aligned} y &= p(a)(p + q) - ap^2 \\ &= ap^2 + apq - ap^2 \\ &= apq \end{aligned}$$

ii. (2 marks)

- ✓ [1] for finding relationship between  $p + q$  and  $a$ .
- ✓ [1] for correct justification of locus.

$$m_{PQ} = \frac{a}{2}$$

Applying gradient formula with  $m = \frac{a}{2}$  between  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$ :

$$\frac{aq^2 - ap^2}{2aq - 2ap} = \frac{a}{2}$$

$$\frac{a(q - p)(q + p)}{2a(q - p)} = \frac{a}{2}$$

$$\therefore p + q = a$$

Substituting into  $x_T$ :

$$x_T = a(p + q) = a^2$$

As the  $x$  coordinate at  $T$  is a constant, the locus of  $T$  is a straight line at  $x = a^2$ .

**Question 12** (Lam)

- Examine  $P(k+1)$ :

(a) (3 marks)

- ✓ [1] for testing base case.
- ✓ [1] for correct use of the inductive hypothesis in the testing.
- ✓ [1] for final answer.

Let  $P(n)$  be the proposition

$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^n \times n}{(n+1)(n+2)} \\ = \frac{2^{n+1}}{n+2} - 1 \end{aligned}$$

for all integers  $n \geq 1$ .

- Base case:  $P(1)$

$$\begin{aligned} \text{Left: } \frac{2 \times 1}{2 \times 3} &= \frac{2}{6} \\ &= \frac{1}{3} \\ \text{Right: } \frac{2^{1+1}}{1+2} - 1 &= \frac{2^2}{3} - 1 \\ &= \frac{4}{3} - \frac{3}{3} \\ &= \frac{1}{3} \end{aligned}$$

Hence  $P(1)$  is true.

- Inductive hypothesis: assume  $P(k)$  is true for  $k \in \mathbb{Z}^+$ , i.e.  $P(k)$  is

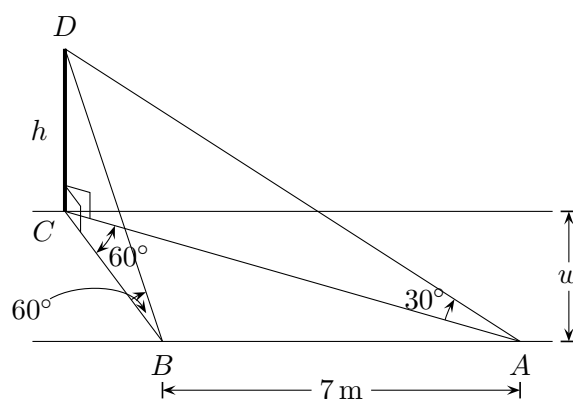
$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^k \times k}{(k+1)(k+2)} \\ = \frac{2^{k+1}}{k+2} - 1 \end{aligned}$$

$$\begin{aligned} \frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} \\ + \cdots + \frac{2^k \times k}{(k+1)(k+2)} \\ + \frac{2^{k+1}(k+1)}{(k+2)(k+3)} \\ = \frac{2^{k+1} \times (k+3)}{k+2} - 1 + \frac{2^{k+1}(k+1)}{(k+2)(k+3)} \\ = \frac{2^{k+1}(k+3) + 2^{k+1}(k+1)}{(k+2)(k+3)} - 1 \\ = \frac{2^{k+1}(2k+4)}{(k+2)(k+3)} - 1 \\ = \frac{2 \times 2^{k+1} \cancel{(k+2)}}{\cancel{(k+2)}(k+3)} - 1 \\ = \frac{2^{(k+1)+1}}{(k+1)+2} - 1 \end{aligned}$$

 $\therefore P(k+1)$  is also true, and  $P(n)$  is true by induction.

(b) i. (3 marks)

- ✓ [1] for both  $CB$  and  $CA$  in terms of  $h$ .
- ✓ [1] for correct usage of cosine rule in  $\triangle ABC$ .
- ✓ [1] for demonstrating  $h = \sqrt{21}$ .



- In  $\triangle DCB$ ,

$$\begin{aligned} \frac{h}{CB} &= \tan 60^\circ = \sqrt{3} \\ \therefore CB &= \frac{h}{\sqrt{3}} \end{aligned}$$

- In  $\triangle DCA$ ,

$$\frac{h}{CA} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\therefore CA = h\sqrt{3}$$

- Applying cosine rule in  $\triangle ABC$ :

$$AB^2 = BC^2 + AC^2 - 2(BC)(AC) \cos 60^\circ \text{ii. (2 marks)}$$

$$7^2 = \frac{h^2}{3} + 3h^2 - 2h^2 \times \frac{1}{2}$$

$$\therefore 49 = \frac{7}{3}h^2$$

$$h^2 = 21$$

$$h = \sqrt{21}$$

- ii. (3 marks)

- ✓ [1] for area by sine rule
- ✓ [1] for area by  $\frac{1}{2} \times h_{\perp} \times \text{base}$
- ✓ [1] for final value of  $w$ .

- Finding the area by base and perpendicular height, where the perpendicular height is denoted  $w$ :

$$A = \frac{1}{2} \times 7 \times w$$

- Finding the area by the sine rule:

$$CB = \frac{\sqrt{21}}{\sqrt{3}} = \sqrt{7}$$

$$CA = \sqrt{21}\sqrt{3} = 3\sqrt{7}$$

$$A = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2}(BC)(AC) \sin 60^\circ$$

$$= \frac{1}{2}(\sqrt{7})(3\sqrt{7}) \frac{\sqrt{3}}{2}$$

Equating areas,

$$\frac{7}{2}w = \frac{1}{2} \times (\sqrt{7})(3\sqrt{7}) \frac{\sqrt{3}}{2}$$

$$\frac{7}{2}w = \frac{7}{4} \times 3\sqrt{3}$$

$$w = \frac{3\sqrt{3}}{2} \text{ m}$$

- (c) i. (2 marks)

$$f(x) = \tan(x^2)$$

Applying chain rule,

$$f'(x) = \sec^2(x^2) \times 2x$$

$$= 2x \sec^2(x^2)$$

$$\int_{-a}^a x \sec^2(x^2) dx = 0$$

( $y = x$  is odd,  $y = \sec^2(x^2)$  is even, i.e. odd  $\times$  even = odd. Integrating an odd function over a balanced interval  $-a < x < a$  produces zero.)

**Alternatively,**

$$\int_{-a}^a x \sec^2(x^2) dx = \frac{1}{2} [\tan(x^2)]_{-a}^a$$

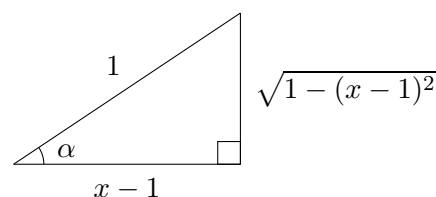
$$= \frac{1}{2} (\tan a^2 - \tan a^2)$$

$$= 0$$

- (d) (2 marks)

- ✓ [1] for relationship between  $\cos \alpha$  and  $x$ .
- ✓ [1] for simplest expression.

Let  $\alpha = \cos^{-1}(x-1)$ . Hence  $\cos \alpha = x-1$ . Draw a right  $\angle$  triangle with adjacent side to  $\alpha$  of length  $x-1$  and unit hypotenuse.



$$\therefore \sin(\cos^{-1}(x-1)) = \sin \alpha$$

$$= \sqrt{1 - (x-1)^2}$$

$$= \sqrt{(1 - (x-1))(1 + (x-1))}$$

$$= \sqrt{(2-x)x}$$

**Question 13** (Lawson)

(a) (2 marks)

$$\begin{aligned}\int \frac{1}{\sqrt{9-4x^2}} dx &= \int \frac{1}{\sqrt{4\left(\frac{9}{4}-x^2\right)}} dx \\ &= \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C\end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct relationship between differentials.
- ✓ [1] for correct primitive.
- ✓ [1] for final answer.

$$\int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx$$

With substitution  $u = 3 + e^x$ :

$$\begin{aligned}\frac{du}{dx} &= e^x \\ \therefore du &= e^x dx \\ x=0 \quad u &= 3 + e^0 = 4 \\ x=\ln 5 \quad u &= 3 + e^{\ln 5} = 8 \\ \int_0^{\ln 5} \frac{e^x}{\sqrt{3+e^x}} dx &= \int_{u=4}^{u=8} \frac{du}{\sqrt{u}} \\ &= \int_4^8 u^{-\frac{1}{2}} du \\ &= \left[2u^{\frac{1}{2}}\right]_4^8 \\ &= 2(\sqrt{8} - \sqrt{4}) \\ &= 2(2\sqrt{2} - 2) \\ &= 4(\sqrt{2} - 1)\end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned}y &= \tan^{-1}(e^x) \\ \frac{dy}{dx} &= \frac{1}{1+(e^x)^2} \times e^x = \frac{e^x}{1+e^{2x}}\end{aligned}$$

ii. (2 marks)

- ✓ [-1] for each newly introduced error.

At  $x = 0$ 

$$\begin{aligned}y &= \tan^{-1} e^0 = \tan^{-1} 1 \\ &= \frac{\pi}{4}\end{aligned}$$

Finding the gradient at  $x = 0$ :

$$\frac{dy}{dx} = \frac{e^0}{1+e^0} = \frac{1}{2}$$

Applying point-gradient formula,

$$\begin{aligned}\frac{y - \frac{\pi}{4}}{x - 0} &= \frac{1}{2} \\ y - \frac{\pi}{4} &= \frac{1}{2}x \\ \therefore y &= \frac{1}{2}x + \frac{\pi}{4}\end{aligned}$$

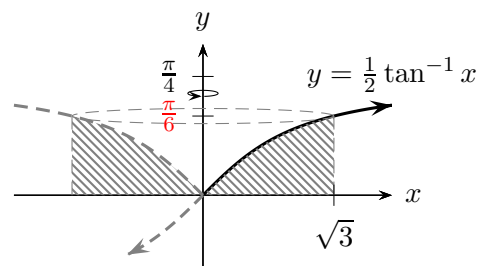
iii. (1 mark) As  $x \rightarrow \infty$ ,  $e^x \rightarrow \infty$ .  
Hence  $y = \tan^{-1}(e^x) \rightarrow \frac{\pi}{2}$ .

(d) i. (2 marks)

$$\begin{aligned}\int \tan^2 2x dx &= \int (\sec^2 2x - 1) dx \\ &= \frac{1}{2} \tan 2x - x + C\end{aligned}$$

ii. (3 marks)

- ✓ [1] for changing subject to  $x$ , and limits of integration along  $y$  axis.
- ✓ [1] for correct primitive
- ✓ [1] for final answer.



When  $x = \sqrt{3}$ ,  $y = \frac{1}{2} \tan^{-1} \sqrt{3} = \frac{\pi}{6}$ .  
Changing subject from  $x$  to  $y$ :

$$\begin{aligned}y &= \frac{1}{2} \tan^{-1} x \\ 2y &= \tan^{-1} x \\ \therefore x &= \tan 2y\end{aligned}$$

Volume by rotating along  $y$  axis:

$$\begin{aligned} V_1 &= \pi \int_a^b x^2 dy \\ &= \pi \int_0^{\frac{\pi}{6}} \tan^2 2y dy \\ &= \pi \left[ \frac{1}{2} \tan 2y - y \right]_0^{\frac{\pi}{6}} \\ &= \pi \left( \frac{1}{2} \tan \frac{\pi}{3} - \frac{\pi}{6} \right) \\ &= \pi \left( \frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \end{aligned}$$

Finding the volume of the cylinder,

$$\begin{aligned} V_{\text{cylinder}} &= \pi r^2 h \\ &= \pi \left( \sqrt{3} \right)^2 \times \frac{\pi}{6} \\ &= \frac{\pi^2}{2} \end{aligned}$$

Volume generated (shaded):

$$\begin{aligned} V &= V_{\text{cylinder}} - V_1 \\ &= \frac{\pi^2}{2} - \left( \pi \left( \frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \right) \\ &= \frac{4\pi^2}{6} - \frac{\pi\sqrt{3}}{2} \\ &= \frac{2\pi^2}{3} - \frac{\pi\sqrt{3}}{2} \end{aligned}$$

### Question 14 (Gan)

(a) i. (2 marks)

✓ [1] for identifying parameters for maximum height.

✓ [1] for finding value of maximum height.

Maximum height occurs when  $\dot{y} = 0$ :

$$\begin{aligned} y &= 30t - 5t^2 \\ \dot{y} &= 30 - 10t = 0 \\ \therefore 10t &= 30 \\ t &= 3 \end{aligned}$$

Substitute into  $y$ :

$$\begin{aligned} y &= 30(3) - 5(3^2) \\ &= 90 - 45 \\ &= 45 \text{ m} \end{aligned}$$

ii. (2 marks)

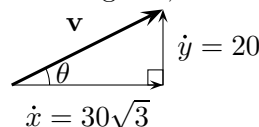
✓ [1] for correct vector diagram

✓ [1] for magnitude and direction of velocity.

When  $t = 1$ :

$$\begin{aligned} \dot{x} &= 30\sqrt{3} \\ \dot{y} &= 30 - 10t \Big|_{t=1} \\ &= 20 \end{aligned}$$

Drawing vector diagram,



$$\begin{aligned} v^2 &= (20)^2 + (30\sqrt{3})^2 \\ &= 400 + 2700 \\ &= 3100 \\ &= 10\sqrt{31} \text{ ms}^{-1} \end{aligned}$$

Finding the angle at  $t = 1$

$$\begin{aligned} \tan \theta &= \frac{20}{30\sqrt{3}} \\ \theta &= 21^\circ 3' \end{aligned}$$

(b) i. (2 marks)

✓ [1] for integrating to obtain  $\frac{1}{2}v^2$

✓ [1] for showing required result.

$$\ddot{x} = -4x + 4$$

Integrating,

$$\begin{aligned} \frac{1}{2}v^2 &= \int (-4x + 4) dx \\ &= -2x^2 + 4x + C_1 \\ v^2 &= -4x^2 + 8x + C_2 \end{aligned}$$

When  $t = 0$ ,  $v^2 = 4 \times 3 = 12$ ,  $x = 0$ :

$$\begin{aligned} 12 &= -4(0) + 8(0) + C_2 \\ \therefore C_2 &= 12 \\ \therefore v^2 &= -4x^2 + 8x + 12 \\ &= -4(x^2 - 2x - 3) \\ &= -4(x - 3)(x + 1) \end{aligned}$$

ii. (2 marks)

- Extremities of motion at  $x = 3$  and  $x = -1$ .
- Centre of motion at  $x = \frac{3-1}{2} = 1$
- Amplitude  $a = 2$

iii. (3 marks)

- ✓ [1] for obtaining completed square form
- ✓ [1] for obtaining an inverse trigonometric primitive
- ✓ [1] for correct equation

Completing the square,

$$\begin{aligned} v^2 &= -4(x^2 - 2x + 1 - 4) \\ &= -4((x-1)^2 - 4) \\ &= 4(4 - (x-1)^2) \\ \therefore v &= \pm 2\sqrt{4 - (x-1)^2} \end{aligned}$$

As  $v$  is non determinant due to initial conditions, assume positive root and perform separation of variables:

$$\begin{aligned} \frac{dx}{dt} &= 2\sqrt{4 - (x-1)^2} \\ \frac{dx}{\sqrt{4 - (x-1)^2}} &= 2 dt \end{aligned}$$

Integrating,

$$\begin{aligned} \int \frac{dx}{\sqrt{4 - (x-1)^2}} &= \int 2 dt \\ \sin^{-1}\left(\frac{x-1}{2}\right) &= 2t + C_3 \end{aligned}$$

When  $t = 0$ ,  $x = 0$ :

$$\begin{aligned} \sin^{-1}\left(-\frac{1}{2}\right) &= C_3 \\ \therefore C_3 &= -\frac{\pi}{6} \\ \frac{x-1}{2} &= \sin\left(2t - \frac{\pi}{6}\right) \\ x &= 1 + 2\sin\left(2t - \frac{\pi}{6}\right) \end{aligned}$$

**Alternatively**, given the initial velocity was not defined in terms of direction, students may have obtained a different phase shift when

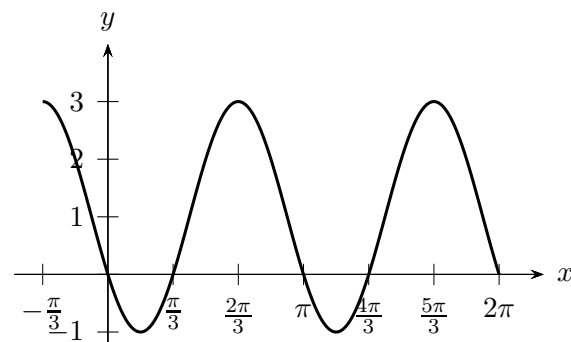
performing the steps above and assumed a negative root. This results in  $x = 2\cos\left(2t - \frac{2\pi}{3}\right) + 1$ .

**Alternatively**, sketch a graph assuming a cosine as the sinusoid with centre of motion at  $x = 1$ , and use initial conditions  $t = 0$ ,  $x = 0$  to find the phase shift:

$$x = 2\cos(2t + \alpha) + 1$$

When  $t = 0$ ,  $x = 0$ :

$$\begin{aligned} 0 &= 2\cos\alpha + 1 \\ \cos\alpha &= -\frac{1}{2} \\ \alpha &= \frac{2\pi}{3} \\ x &= 2\cos\left(2t + \frac{2\pi}{3}\right) + 1 \end{aligned}$$

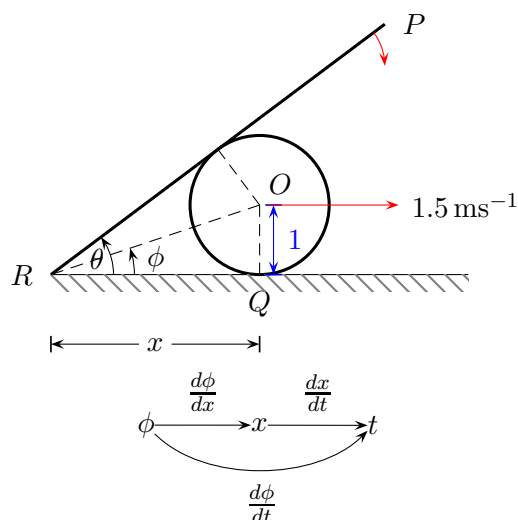


(c) i. (1 mark)

- $OQ = 1$
- $\tan\phi = \frac{1}{x}$

ii. (3 marks)

- ✓ [1] correct derivative  $\frac{d\phi}{dx}$ .
- ✓ [1] for finding  $\frac{d\phi}{dx} = -\frac{1}{37}$ .
- ✓ [1] for finding  $\frac{d\theta}{dt}$ .



Applying chain rule,

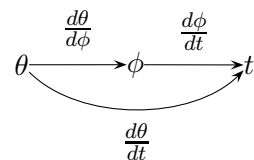
$$\frac{d\phi}{dt} = \frac{d\phi}{dx} \times \frac{dx}{dt}$$

Given  $\tan \phi = \frac{1}{x}$ , rearranging to find  $\frac{d\phi}{dx}$ :

$$\begin{aligned}\phi &= \tan^{-1}(x^{-1}) \\ \frac{d\phi}{dx} &= \frac{1}{1 + (x^{-1})^2} \times -x^{-2} \\ &= \frac{1}{1 + \frac{1}{x^2}} \times \frac{-1}{x^2} = -\frac{1}{x^2 + 1} \Big|_{x=6} \\ &= -\frac{1}{37}\end{aligned}$$

Evaluating  $\frac{d\phi}{dt}$ :

$$\frac{d\phi}{dt} = -\frac{1}{37} \times \frac{3}{2} = -\frac{3}{2 \times 37}$$



Using the  $\theta - \phi$  chain,

$$\theta = 2\phi$$

$$\therefore \frac{d\theta}{d\phi} = 2$$

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{d\theta}{d\phi} \times \frac{d\phi}{dt} \\ &= 2 \times -\frac{3}{2 \times 37} \\ &= -\frac{3}{37} \text{ rad/s}\end{aligned}$$